Homework Assignment 1

Math 292, Spring 2026

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February 6, 2026

Problem 1:

Subtract the min required 12 balls from the initial 35, leaving 23 balls remaining. We then distribute these using stars and bars:

$$\binom{28}{5} = 98280 \text{ ways}$$

Problem 2:

$$\frac{\binom{13}{1}\binom{4}{3}\binom{12}{1}\binom{4}{2}}{\binom{52}{5}} \approx 0.00114$$

Problem 3:

Let C_1 = event that the two-headed coin is picked, C_2 = event that the fair coin is picked, and C_3 = event the biased coin is picked, finally, let H = event that heads is flipped.

Thus, by using Bayes theorem, we have:

$$\begin{aligned} P(C_1|H) &= \frac{P(H|C_1) * P(C_1)}{P(H|C_1) * P(C_1) + P(H|C_2) * P(C_2) + P(H|C_3) * P(C_3)} \\ &= \frac{\frac{1}{3}}{\frac{1}{3} + (\frac{1}{2})(\frac{1}{3}) + (\frac{1}{3})(\frac{3}{4})} \\ &= \frac{4}{9} \end{aligned}$$

Problem 4: Using the inclusion-exclusion principle, we have that

$$P(\text{void bridge hand in at least one suit}) = \frac{\binom{4}{1}\binom{39}{13}}{\binom{52}{13}} - \frac{\binom{4}{2}\binom{26}{13}}{\binom{52}{13}} + \frac{\binom{4}{3}\binom{13}{13}}{\binom{52}{13}} \approx 0.0511$$

Problem 5:

We have P(dog) = .36, P(cat) = .30, and P(cat | dog) = 0.22

(a)

$$\begin{aligned} \text{P(cat)} \cap \text{P(dog)} &= \text{P(cat} \mid \text{dog)} * \text{P(dog)} \\ &= 0.22 * 0.36 \\ &= 0.0792 \\ &= 7.92\% \end{aligned}$$

(b)

$$\begin{aligned} \mathrm{P}(\text{dog} \mid \text{cat}) &= \frac{\mathrm{P}(\text{dog}) \cap \mathrm{P}(\text{cat})}{\mathrm{P}(\text{cat})} \\ &= \frac{0.0792}{0.3} \\ &= 0.264 \\ &= 26.4\% \end{aligned}$$

Problem 6:

Theres a total of $5 + 6 + 8 = 19$ balls

(a)

$$\frac{\binom{5}{3} + \binom{6}{3} + \binom{8}{3}}{\binom{19}{3}} = \frac{86}{969} \approx 0.089$$

(b)

$$\frac{\binom{5}{1} * \binom{6}{1} * \binom{8}{1}}{\binom{19}{3}} = \frac{240}{969} \approx 0.248$$

With replacement:

(a)

$$(\frac{5}{19})^3 + (\frac{6}{19})^3 + (\frac{8}{19})^3 = \frac{853}{6859} \approx 0.124$$

(b)

$$3! * (\frac{5}{19} * \frac{6}{19} * \frac{8}{19}) = \frac{1440}{6859} \approx 0.21$$